

Puzzles

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This document is a collection of interesting puzzles that I find that I started on the 12th of September, 2024. For each problem I give some background as to where I found it and its difficulty, then give some hints. The solution is then presented, and I give my thoughts on the problem, extensions, and other interesting points of relevance. It is split into five sections in decreasing order of how abstract the area is, and within each section it is organised chronologically by when it was added (and by proxy, when I found/solved the problem). This document is available at: <https://github.com/HenryGinn/Puzzles>.

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1 LOGIC

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1.1 The Twelve Islanders Problem

This is a classic problem. I first came across this problem when My Oxford admissions tutor, Jeremy, gave it to me in 2016. There he posed it as doing it in as few steps as possible rather than the version presented here where it is given that it is possible with three. I next encountered it on an episode of Brooklyn 99 which I imagine played a large part in the popularisation of the problem. I solved it after my second watch through in 2024 and it took me an hour. Chat GPT version 4o is unable to solve it, even with prodding, and I think it is actually quite hard. Jeremy said very few people had managed to solve it, and I rate it as 6 out of 10 in terms of difficulty. This is a good problem to talk about on a walk with others, although a pen and paper is beneficial.

There are twelve people on an island. Eleven of them have the exact same weight, but the last one is a different weight, either lighter or heavier. There is a set of scales on the island that can be used only three times. There is no limit on how many people can be on either side of the scales, and it will either tilt left, right, or stay balanced. The problem is to find a strategy where the odd islander can be identified, and whether they are heavier or lighter than the others.

Hints:

1. Before diving in to testing weighings, try think about how much information you need to extract and upper bounds on how much you can extract.
2. In a single weighing there are three outcomes, in two weighings there are nine, and in three weighings there are 27. There are twelve possibilities for who the odd one out is and as they can be lighter or heavier there are 24 possibilities that a winning strategy needs to distinguish between. This means almost all information needs to be extracted at each stage.
3. Weighing five vs five islanders and having it not balance leaves ten remaining possibilities yet only nine possibilities can be distinguished in the two remaining weighings.
4. The first weighing is comparing four islanders against another four islanders.
5. After the first weighing, there will be at least four islanders that are ruled out. The islanders which have been determined to not be the odd ones out can be used in future weighings.

Solution:

As mentioned in the hints, the most important part is extracting as much information as possible. In the below discussion we will write a possibility as an islander number followed by an “H” or an “L” for “Heavier” or “Lighter” respectively, such as “1H”. We will refer to an islander that has been ruled out from being the odd one out as known-weight islander and denote them with a “K”. For convenience the word “potentially” has been dropped when referring to potentially heavy and potentially light islanders, and we will denote them by “H” and “L” respectively.

Before we begin, we make clear one of the ideas made in the hints about how much information we can extract. In a single weighing there are three outcomes that could happen: down on the left, balanced, and down on the right. No matter what combination of islanders are on the scales, we know it will be one of those outcomes only. For each of these outcomes we could do a second weighing, meaning there are $3 \cdot 3 = 9$ outcomes we could distinguish between. Repeating this reasoning again we see there are $3^3 = 27$ outcomes for three weighings.

How do we use this information? Suppose we have decided on some strategy and we follow it for two sets of islanders where the odd one out is different between the groups, let us say 1H and 3L. For the sake of example, suppose for both sets of islanders we witness the scales balance, then go down on the left, then down on the left again in both cases. How could we tell which group had islander 1H and which group had 3L? This demonstrates how any working strategy would need to ensure that each possibility for the odd one out ends up with a different sequence of weighing results.

For the first weighing, all the islanders are indistinguishable as we have zero knowledge that sets any apart from the rest, therefore the only decision is how many to put on each side. The first deduction made is that to gain any information at all, there must be an equal number of people on each side of the scales. This can be seen to be true at any step in the process, even if we know some information about the islanders.

Many people's first thought is to have six islanders on either side. It cannot stay level as the odd one out must be on one of the sides. This is already enough to rule out this option as we cannot gain any information from the scales balancing making us too inefficient. Suppose islanders 1 to 6 and 7 to 12 are put on the left and right respectively and without generality the scales tip down on the left. The remaining possibilities are 1H, 2H, 3H, 4H, 5H, 6H, 7L, 8L, 9L, 10L, 11L, and 12L which is twelve possibilities. We can only distinguish between nine possibilities in two weighings, so this will not work.

A near identical argument can be made if five people are put on either side of the scales. If they do not balance then there are ten states remaining which is more than nine, and thus impossible. Alternatively we could argue that if it balances then too much information has been gained about the two islanders left over. There are four possibilities: if the first ten were on the scales these would be 11L, 11H, 12L, and 12H. As we can distinguish between nine possibilities with the remaining weighings we are wasting five possibilities. We can only afford to waste $27 - 24 = 3$ possibilities, therefore a contradiction.

If three islanders are put on either side of the scales and the scales balance then the problem has been reduced to determining the odd one out among six islanders in two weighings. This has 12 different states which is too many, and therefore the problem is impossible. We can conclude that four islanders are needed on either side of the scales for the first weighing, and the problem now splits into two cases.

- Case A: the islanders balance. The odd one out is among the four islanders not on the scales, and no information is known about whether they are heavier or lighter. We name these islanders 1, 2, 3, and 4 and there are eight possibilities to distinguish between. We call the islanders not on the scales 5 to 12. This case is easier to analyse.
- Case B: the islanders do not balance and without loss of generality we say it goes down on the left. The odd one out is one of the eight islanders on the scales which we will name 1 to 8, with the islanders not on the scales named 9 to 12. We know the odd one out is one of 1H, 2H, 3H, 4H, 5L, 6L, 7L, or 8L, again 8 possibilities.

Starting with case A, we again need to determine how many people belong on the scales. If one islander is put on each side of the scales and they happen to balance then there are at least two islanders where we know no information about them. This leaves four possibilities in one weighing which is impossible to distinguish between.

There are three ways to weigh two vs two islanders: having zero, one, or two known-weight islanders on the left side. The first of these is similar to six vs six for the first weighing. We could be left with either 1H, 2H, 3L, and 4L or 1L, 2L, 3H, and 4H, too many to distinguish with the last weighing. The last of these is ruled out for the same reasoning as one vs one - it leaves two unknown-weight islanders uninvolved so we may not gain any knowledge about them.

Weighing one known-weight islander and an unknown-weight islander against two unknown-weight islanders does work however. Without loss of generality we can do 1 and K against 2 and 3. If it balances then islander 4 is the odd one out and we can weigh them against a single known-weight islander. If it goes down on one side, without loss of generality let us say the left, we have 1H, 2L, and 3L as possibilities. Weighing 2L against 3L is the last weighing with the outcomes listed below.

- Down on the left: 3L is the odd one out
- Balances: 1H is the odd one out
- Down on the right: 2L is the odd one out

For the sake of completeness, we consider the other possible second weighings for case A using three or four islanders on each side. We observe that if we put a known-weight islanders on each side then they will cancel each other out. These weighings will give us identical information to the weighings where they were never added so we do not need to consider those weighings. The remaining cases that are not equivalent to each other or a previous strategy are listed below.

1. 1 2 3 vs K K K: This is easily seen to work.
2. 1 2 3 vs 4 K K: This does not work. If the scales go down on the left then the possibilities are 1H, 2H, 3H, and 4L.
3. 1 2 3 4 vs K K K K: This does not work. If the scales go down on the left then the possibilities are 1H, 2H, 3H, and 4H.

For case B, we start by making an observation we learnt from the case A analysis. If we have narrowed the possibilities down to three islanders that were previously involved in a weighing then we can determine the odd one out in one more weighing. The four cases are L L L, L L H, L H H, and H H H, and the odd one out can be determined by weighing L vs L or H vs H. We are using all the information we can gain from one weighing when we do this and all the unknown islanders were involved in the first weighing. From this we deduce that we must eliminate five unknown islanders in the second weighing.

This motivates the following strategy. Take any three of the islanders 1 to 8 and leave them off the scales for the second weighing. We know from the above observation that if we eliminate the other islanders from being the odd one out then we can determine which of the three are the odd one out. This leaves five islanders of unknown weight that must be included in the second weighing. There are two distinctly different cases as swapping the roles of heavy and light is equivalent to swapping the roles of left and right which are H H H H L and H H H L L.

When the scales balance we know the odd one out has been left off the scales and we are done. It remains to only consider cases where the scales do not balance. For each side that goes down there can only be at most three islanders potentially responsible for that as otherwise we would not be able to deduce the odd one out. This means the number of heavy islanders on the left plus the number of light islanders on the right cannot be more than three, and similarly when swapping left and right. This is a necessary and sufficient condition¹ and it can be checked that only four distinctly different configurations work: HHL vs HHK, HHHL vs HKKK, HHL vs HLK, HHLL vs HKKK.

We return to the decision to leave only three islanders off the scales at the second weighing. If we had left four islanders off and the scales balanced then we could not determine the odd one out with the last weighing. If we had two islanders off then we can in fact find more solutions however: HHL vs HHL and HHHLL vs HKKKK. Enumerating all cases proves that these are all the solutions, meaning there are $2 \cdot 6 = 12$ different strategies that work.

Extensions and Comments:

We note that we found an upper bound of 27 different states we could distinguish between which means we cannot immediately rule out the possibility of distinguishing the odd one out between thirteen islanders as $13 \cdot 2 = 26 < 27$. By the same analysis we did earlier we reach the same conclusion that the first weighing is four vs four islanders, leaving five islanders to distinguish between if the scales balance. There are now ten states to distinguish between instead of eight, breaking our bound of nine states in two weighings, thus we conclude that the problem is impossible with thirteen islanders.

¹We note that HHHLL vs KKKKK satisfies these conditions but has to be ruled out because we do not have five known-weight islanders.

1.2 Prisoners and the Lever

Frustratingly I cannot remember the source for this problem, although I think I came across it around 18 as an interview problem from a YouTube video. This is a good problem to talk about on a walk as it does not benefit from the use of pen and paper. I rate the easy version of this problem is a 4/10 difficulty, and the harder versions both at 7/10. The distinctions given in version 2a and 2b can be dropped for simplicity, although I think they are worth including.

There are $n + 1$ prisoners who are each going to be put in their own individual cells. One at a time, in no particular order, each prisoner will be brought into a room that has a lever in, and they have a choice of whether to change its state or not. The prisoner will then be sent back into their cell and the next prisoner brought in. This process repeats until one of the prisoners announces that all prisoners have been in the room. If they are correct then all prisoners are free to go, although if they are wrong then they are all imprisoned forever/sentenced to death. The prisoners are allowed to communicate freely before being put into their cells to agree a strategy, and the problem is to find a strategy that guarantees all prisoners go free.

The order that the prisoners enter the room does not necessarily repeat, any prisoner could be brought into the room at any point. A constraint that I believe to be necessary for this problem to be well-posed is that all prisoners will visit the room an infinite number of times. This will be referred to as the recurrent constraint. This becomes important when evaluating whether a solution is robust to a malicious actor who chooses the order (from now on referred to as a malicious chooser) and determining the probability of failure. The recurrent constraint can be assumed at all times, and any malicious chooser needs to ensure it holds as well. There are multiple variations of this problem, and hints for part 2 will assume familiarity with part 1 (and thus includes spoilers).

1. (Easy version) The lever is known to start in the up position.
2. (Hard version) The position of the lever is not known.
 - (a) (Robust) The proposed strategy must work even if a malicious chooser decides the order that the prisoners enter the room. This condition is defined as “Robust”.
 - (b) (Initial state) The starting state of the lever must be determined as well. Assuming a random order, this must work with probability 1. This condition is defined as “Certain”.

Hints for part one:

1. The announcement that all prisoners have visited the room does not need to happen at the earliest possible moment. The only requirement is that all prisoners have been in the room at least once, meaning all prisoners may have visited the room multiple times when the announcement is made.
2. Whether any individual prisoner has visited the room is not important, only the total number
3. Only one prisoner needs to be able to make the claim that all have visited the room.
4. If a prisoner flicks the lever and another prisoner flicks the lever again before the prisoner making the claim enters the room, this state will be indistinguishable to where no prisoner flicked a lever.

Solution for part one:

This strategy works via communicating all information to a special prisoner who will be referred to as the communicator. All other prisoners will be referred to simply as prisoners. The key is to ensure that the communicator counts each prisoner at least once, but no more. Each prisoner follows a simple rule: if the lever is up and they have not flicked it before, flick the lever down, but otherwise do nothing. If the communicator enters the room and sees the lever up, they do nothing, but if they see the lever down, they gain information.

The lever being down means a prisoner who had not flicked the lever before has flicked the lever. The communicator can flick the lever up to reset it back into the original state, and they add one to the count of prisoners who have flicked the lever. As no prisoner can flick the lever twice, no prisoner will get double counted. As no one can flick the lever up other than the communicator, no prisoner can get missed. As the communicator will know how many prisoners there are due to being allowed to communicate beforehand, they will know when all prisoners have been in the room when they have counted to n . The existence of the communicator is why this problem is defined with $n + 1$, not n .

General hints for part 2:

1. The part one strategy does not work for part 2. If the communicator enters the room first and sees the lever down, they will not know whether this is because they are the first in the room or if another prisoner has flicked it down. Their count could be one off at the end, and they would not know if it is because the lever started down or they have not waited long enough.
2. The main idea from the part one of counting prisoners by flicking the lever down can be extended.

Hints for part 2a:

1. Consider letting prisoners flick the lever multiple times.
2. Use the pigeonhole principle.

Hints for part 2b:

1. Determine first how it is possible for the communicator to learn the initial lever state.
2. If the prisoners only flick the lever down, then when the lever is down the communicator can know whether this is due to them or a prisoner on all but their first visit to the room.

Solution for part 2a:

This solution works using the pigeonhole principle. The strategy for each prisoner is updated so that they will flick the lever down if they have never flicked it, or have only flicked it one time. If the lever starts in the up state, the communicator will count the lever being in the down state $2n$ times when all n prisoners have flicked the lever down two times each. If the lever started in the down position however, the communicator can count the lever being down $2n$ times like before, but the prisoners would have only flicked the lever down $2n - 1$ times collectively. This is because the first time the communicator entered the room the lever had not been flicked down by any prisoner, yet it was counted anyway. If there were only $n - 1$ prisoners, they could only flick the lever down collectively $2n - 2$ times. Even if the lever started in the down position, the most the communicator could count to would be $2n - 1$, making it impossible for the communicator to count the lever being in the down position $2n$ times. By the pigeonhole principle, each of the n prisoners must have been in the room at least once, and in fact at most one prisoner could not have been in the room twice.

Solution to part 2b:

For the communicator to determine the lever starting position, it is easiest if no touches the lever before the communicator enters the room. To see this, consider the following two situations. In situation A, the communicator is the first in the room, and in situation B, a prisoner flicks the lever and then the communicator enters the room. The communicator would not be able to distinguish between these situations, yet any strategy where the lever is flicked before the communicator enters the room would need to account for this situation happening. It is potentially possible that the prisoner who flicked the lever could somehow communicate this information to the communicator, and while I have not ruled this out, I am highly skeptical that such a strategy exists. If this communication is indeed not possible then no one touching the lever before the communicator is a necessary part of the strategy.

For the lever not to be touched before the communicator enters the room, the strategy for the prisoners must involve not touching the lever until they receive some sort of signal from the communicator. The communicator will need to add extra flips to communicate to the prisoners that they are allowed to flip the lever. The communicator knows how many times they have flipped the lever up and down, therefore this does not throw off their count. The rule for the prisoners is that once they have seen the lever in both the up and down state, the lever is in the up state, and they have not flicked it before, then flick it down. This is the same as before, but with addition of receiving a signal.

The rule for the communicator is more complicated. If they leave the room with the lever flicked up and the next time they enter they find it down, then they add one to their count. They will not announce until they have counted to n , and by the logic of easier version this is a guarantee that all prisoners have been in the room. The complication is that they will need to sometimes pull the lever down themselves. If they have left the lever up for a long time and it is not getting flicked down, then it would indicate that a prisoner has only seen the lever in the up position, and thus has not received the signal to pull the lever down. The communicator would flick the lever down and leave it for a while to give the other prisoners an opportunity to see it, and then flick it back up to give them an opportunity to flick it down.

Extensions and Comments:

I like this problem as initially it seems impossible due to how little information it appears can be communicated through the lever, especially when the order is unknown. After solving the first part, it seems strange that the second part is possible given that the information the first part relied upon has been taken away. An equivalent formulation of the version 2a is where there are two levers in unknown starting positions, but all prisoners must flick a lever exactly once when entering the room. The part one strategy is modified by selecting one lever to be the important one, and the other lever to be flicked when the first lever should not be flicked. It surprised me that this problem involved one of the prisoners being special, and that there was not some symmetrical solution where each prisoner was on a level footing.

The proposed strategy for 2b is not robust to malicious actors. If a malicious actor were choosing the order, it is possible to make it so that only one of the prisoners gets the signal to flick the lever down, and thus the game never terminates. If the prisoner order was random however, then there is a non-zero chance, p , that a given prisoner gets the signal that they can flick the lever. As $1 - (1 - p)^n \rightarrow 0$ as $n \rightarrow \infty$, the chance that the game goes on forever without the prisoner getting the signal is 0. There are a finite number of prisoners, therefore there is a probability of 1 that every prisoner gets the signal.

A common solution to this problem is for each of the prisoners to wait for a really really long time and then guess that all prisoners must have surely visited the room in that time. This fails the malicious actor test, as a malicious chooser knows the prisoners have to make an announcement after some amount of time. If one of the prisoners announces after entering the room 10,000 times and one of the prisoners had not entered the room once. The prisoners could complain that the malicious chooser broke the recurrent constraint, but they could respond by saying they planned to put the last prisoner in the room the next time. The prisoners strategy means they must make an announcement, and the game will always terminate in a finite amount of time. The malicious chooser will always have the defense that the prisoners just had not waited long enough.

The solution to versions 1 and 2a are not susceptible to the same attack however. There is no limit on the amount of time the the communicator can wait until they make an announcement. If a game continued to infinity with the malicious chooser refusing to put at least one of the prisoners in the room after some point, then the malicious chooser would have broken the recurrent constraint. The “wait a long time and guess” approach fails because the games cannot be infinite in length. If there are k prisoners who have not flicked the lever after t room visits, then the malicious chooser must put at least one of these prisoners in the room after this point due to the recurrent constraint. This will occur after

a finite amount of time, and would lead to one of the k prisoners flicking the lever. The same principle applies to the communicator, and this repeats by induction until all prisoners have been in the room.

The “wait a long time and guess” approach is also not certain. The probability tends to zero as the number of room visits tends to infinity, but it only reaches 0 in the limit. After a finite number of room visits the probability is strictly greater than 0, and thus is not certain. In fact this argument is sufficient to show the method is not robust as robustness implies certainty. To prove this, consider the contrapositive - if a method is not certain, it is also not robust. If a method is not certain then there must exist some sequence of room visits that causes an incorrect announcement to be made. A malicious chooser could give the same sequence, therefore making the method not robust.

I do not know if it is possible to determine the lever starting state with a method that is completely robust to a malicious chooser, but I suspect it is not. There are two types of strategy to consider - those where prisoners can flick the lever before the communicator, and those where they cannot. The proposed strategy to 2b is of the latter type, and must involve some sort of signal that the prisoners can start flicking the lever (they must flick at least once in order to communicate information). The flaw in the given strategy is that there are reasons for the communicator leaving the lever either up or down depending on what type of prisoner they want to see it. A malicious chooser can exploit this to guarantee no prisoner gets their intended signal, and gets around the recurrent condition as they know the communicator has to change the lever state sometimes.

The former type of strategy seems to have a lot less control and has two main complications. The first person to enter the room will be unaware they are the first and once they have determined this some way, they need to communicate the lever state to the communicator. As this method aims to avoid having a signal to start the game, this strategy cannot interfere with original goal of determining when all prisoners have entered the room. I do not believe that the signalling strategies previously discussed would work, and if there is a robust strategy then it is of this type. Nonetheless, I would be very surprised if a strategy of the former type existed, let alone one that was robust.

1.3 The Snails on a Plank Problem

This problem appears very difficult at first, but has a very simple solution that I personally think is rather unintuitive. I've asked several people this problem as of writing this, and all of them fell right into the trap which pulled them away from the true solution. I cannot remember where I got this problem from, and rate this problem 4/10 in terms of difficulty. This is too high a rating given the simplicity of the solution, but it really seems to throw people off, even when the solution is explained to them.

There are n snails on a plank which can be assumed to be one dimensional. The snails are randomly distributed on the plank and can be approximated at point particles. They all move at the same speed but in random directions, either left or right along the plank. When they get to the end they fall off, and they swap directions when running into another snail. It takes a single snail one minute to go from one end of the plank to another. How long does it take for all snails to fall off the plank, and more importantly why is that the answer? Extension: if all left-going snails start to the right of the right-going snails, how many times do snails run into each other?

Hints:

1. The answer is independent of the number of snails and the starting configuration.
2. It takes at most one minute before all snails have fallen off the plank.
3. Consider the case of just two snails, but do not attempt to prove this by induction.
4. Suppose a blue snail heading right collides with a red snail heading left. Both swap directions, but suppose they also swap colours. Visualise what this would look like.

Solution:

As it may be suspected by the complete lack of information given, all the apparent complexity cancels out in some way and the answer is a constant one minute which most people can guess. The trick to understanding why is to look at what happens when two snails interact. Two snails hitting and instantly changing direction is indistinguishable from two snails phasing through each other. This can be demonstrated with two fingers, in one case where they move towards each other and pass by each other, and in the other case where they change directions. The fourth hint suggests this very strongly and is another good way to picture why this is the case.

Applying the reasoning of two snails interacting to all the snails, it would look like all the snails heading right do so unimpeded, and the same for the snails heading left. They simply phase through each other and can therefore be treated individually, thus reducing to the case for one snail which takes at most one minute. Using this, the general case of the extension can be easily seen as the sum of the number of left-going snails to the right of each right-going snail. Where all right-going snails start to the left of the left-going snails, this is the product of the left-going and right-going snails.

Extensions and Comments:

What I believe causes difficulty in this problem is thinking about the snails as individuals. When considering a pair, the answer is almost immediate, but it seems most natural to think of one snail at a time. My dad attempted to program this and I told him one of two things would happen. He would either finish programming it and think there was a bug when it first ran (i.e. expecting to see snails bounding off each other but actually seeing them phase through each other), or he would realise the solution before finishing (when coding a snail collision). The latter of these is exactly what happened.

Many people start off with one and then two snails, but make the mistake of considering three or more. A commonly considered scenario would be to place one snail at the end and another halfway along, and note the snails would travel $0.25 + 0.25 = 0.5$ and $0.25 + 0.75 = 1$ times the length of the plank

respectively. This convinced people it was one minute, but took them no further. I made a Desmos simulating this that shows the behaviour: <https://www.desmos.com/calculator/g2mcmsmc1m>.

Interestingly, it is impossible for two snails to collide after 30 seconds, or equivalently after any snail had travelled a total distance of halfway along the plank. This is a consequence of the fact that the snails must fall off by the end of the minute. Explicitly, a snail has to be within $1 - t$ of the end of the plank it would eventually fall off at time t , where t is the time in minutes. If two snails were on the left side after 30 seconds, then they cannot reach the right side. They must be travelling in opposite directions in order to collide, meaning one would go left and the other would go right. The right-going snail would not reach the right side in time. They cannot be saved by colliding with a left-going snail as this simply kicks the snail down the road - the other snail would take their place.

The problem is phrased rather imprecisely, where the exact question that is sought is actually the supremum of times it takes for snails to fall off. We can go further than this however, and find the probability distribution of times for the last snail to fall off. Let T_i be the time taken for the i^{th} snail to fall off, and $T = \max_i T_i$ be the time for the last snail to fall off. The labelling of the snails does not matter as we are only interested in T , and we can focus on the equivalent situation where snails phase through each other. The time it takes for the snails to reach the end is exactly proportional to the distance from the end they will fall off. As the snails are uniformly distributed in space, the T_i are also uniformly distributed, giving $T_i \sim U(0, 1)$ as a set of independent and identically distributed random variables. The cumulative distribution of T can be found as follows.

$$\begin{aligned}
 \mathbb{P}(T \leq t) &= 1 - \mathbb{P}(T \geq t) \\
 &= 1 - \prod_{i=1}^n \mathbb{P}(T_i \geq t) \\
 &= 1 - \prod_{i=1}^n (1 - \mathbb{P}(T_i \leq t)) \\
 &= 1 - \prod_{i=1}^n (1 - t) \\
 &= 1 - (1 - t)^n
 \end{aligned}$$

1.4 Unpaired Corners on Double Move Rubik's Cube

Hints:

1.

Solution:

Extensions and Comments:

1.5 The Josephus Problem

I found this problem on Numberphile in 2016. This is a good problem to talk about on a walk, although a pen and paper can be beneficial. I gave this problem while on a walk on a holiday to Dublin with friends and together they figured it out in about 10 minutes, including the extension (this has been listed as a hint). This problem does not need advanced maths or logic, and trying out some situations and seeing what happens gives a good entry into the problem which makes it a nice problem solving exercise. I give this problem a difficulty rating of 3/10.

You are part of a group of n knights who have decided to commit ritual suicide in the following way. All the knights stand in the circle, and the starting knight kills the person to their left, going clockwise. The next surviving knight does the same, and this repeats until there is one knight left who would then kill themselves. You have decided you do not fancy killing yourself however, and want to place yourself in the circle so that you are the last surviving knight, and no other knights will find out you did not kill yourself. If the knights are numbered from 1 to n going clockwise, what position should you stand in to survive?

Hints:

1. Run the game through for several small cases ($n < 10$). Some values of n will lead to easier cases than others - make a conjecture about this and prove it.
2. The easy cases are when n is a power of 2.
3. How can you reduce the case of an arbitrary n to a case involving a power of 2?
4. Extension: find a nice representation of the solution in binary.

Solution:

When $n = 2^k$, knight 1 kills 2, knight 3 kills 4, and so on until a loop is complete and it is the turn of knight 1 to kill again. All of the even numbered knights would be dead, and if they were renumbered then the situation would be indistinguishable from starting with $n = 2^{k-1}$. This means that if n is a power of 2, you should go in the first position.

If $n = 2^k + m$ where $0 \leq m < 2^k$ then after m knights have been killed the problem has been reduced to the case where n is a power of 2. This case has been solved already, so it suffices to find the knight that starts the $n = 2^k$ game. There is one killer knight for each killed knight which means if m knights are killed then knight $2m + 1$ is the knight of interest. For this to be well-defined we need to have not wrapped around the full circle which means we need $2m + 1 \leq n$. This will be the case as $2m + 1 = (m + 1) + m \leq 2^k + m = n$.

If $b_k b_{k-1} \dots b_1 b_0$ is the binary expansion of n then $b_{k-1} b_{k-2} \dots b_1 b_0$ is the binary expansion of m . Multiplying m by 2 is a bit shift to the left, and adding 1 gives the binary expansion of $2m + 1$ as $b_{k-1} b_{k-2} \dots b_1 b_0 1$. This is the same as if we took the leading 1 in the binary expansion of n and moved it to the end of the number.

1.6 The Three Gods Riddle

Hints:

1.

Solution:

Extensions and Comments:

1.7 The Three Hats Problem

I came across this problem from a YouTube video by one of my undergraduate friends, JPiMaths. I failed to solve this and gave up too soon. I attempted a brute force approach and in hindsight this was a bad idea, this problem lends itself better to thinking about it rather than programming it. When giving this problem I think coming up with the idea of how to find a strategy rather than explicitly finding a strategy is more important. I give this problem a 5/10 in difficulty, and a pen and paper, or even a spreadsheet would help.

There are three people who are given hats that could be either red, blue, or yellow. The hats could all be the same colour, all different colours, or two the same colour and the third a different colour. They cannot see their own hats, and they cannot communicate to the others. Simultaneously, they will have to guess the colour of their own hat, and they win the game if at least one of them guesses correctly. They are allowed to come up with a strategy before the game starts - what should their strategy be?

Hints:

1. Think about what information each person has available to them.
2. There are $3^3 = 27$ combinations of hats - this is the number of scenarios that their strategy will collectively need to work for.
3. Each person can see two other people who each could be wearing three different hats - this means each person can distinguish $3^2 = 9$ states.
4. Across the three of them they can potentially combine to distinguish between a maximum of $3 \cdot 9 = 27$ scenarios. How can this be achieved?
5. For each possible scenario, only person can guess correctly as otherwise there is not enough information to have at least one person be correct in all other scenarios.

Solution:

The only information available is what colour hat the other two people are wearing for each of the three people. Each person could be looking at nine different combinations, and each of these will correspond to a guess. For a given person, we can take an ordered pair (the two colours the person is looking at) and the colour that the person guesses, and map it to the scenario where it is correct. This function is well-defined because the scenario being mapped to is unique. The ordered pair fixes what the other two could be wearing, and the guess fixes what the given person is wearing.

As each person sees nine distinguishable states, they can only be correct in nine scenarios when using a deterministic strategy. If any two people are both correct for the same scenario, then together they would not be able to cover all 27 scenarios. The question now turns to how to construct such a map for each person so that all 27 scenarios are covered (note that at this stage we have not shown that this is possible). We can think of this as a lookup that takes in a person, the hat colour of the person to their left, and the hat colour of the person to their right, and gives a scenario. The problem is finding a surjective lookup of this form.

Extensions and Comments:

There are 3^{27} possible lookups to search through which sounds intimidating, but it is in fact possible to brute force it manually at this stage. Using a spreadsheet to compute which scenarios are guessed correctly for a given map, we can recurse through the space, rejecting an entry to the lookup if it does not cover a new scenario. This turns out to prune the tree very efficiently. I did not even need to backtrack when finding a solution this way, and my Python code finds all solutions in 3 seconds. It is possible to be more analytical about this however.

If all three are wearing the same colour hat, say blue, then all are looking at blue hats. We know that one of them must guess blue as otherwise this scenario would not be covered, but if all guess blue then more than one would guess correctly. Therefore for the pairs (red, red), (blue, blue), and (yellow, yellow), only one person can guess red, blue, and yellow respectively. In particular this shows that each person must have a different map for the 9 states.

A person cannot guess the same thing for any triple of pairs where either the first in the pair or the second in the pair are all the same. Consider what happens when person 1 always guesses red when they see person 2 and person 3 wearing either (red, red), (red, blue), or (red, yellow). If person 3 is looking at person 1 wearing red and person 2 wearing red, then it is impossible for them to guess anything. The scenarios (red, red, red), (red, red, blue), and (red, red, yellow) have already been covered by person 1. Person 3 sees (red, red) in all three cases, and so cannot map (red, red) to anything new. The reasoning is similar for the other cases.

It is also the case that each person guesses three of each colour. I have not been able to prove this however, or verify whether it occurs for any other non-trivial number of people and colours. There are $10752 = 6 \cdot 2^8 \cdot 7$ lookups that work for the $n = 3$ case. Being a multiple of 6 is not a surprise as any permutation of the three colours for a given solution will also work as a solution. I suspect there is a reason why such a large power of 2 is a factor and why 7 appears, but I have not been able to make progress on this.

1.8 The Infinite Hats Problem

This puzzle frustrated me when I first heard it as I knew I would not be figuring it out. It appears completely impossible, and this is the reaction of everyone I have shared it with, and I imagine whoever came up with it did so as a demonstration of the weird results implied by the axiom of choice. I do not think it is fair to give this puzzle to a non-mathematician with the expectation that they could solve it as only a mathematician would think of using the machinery needed to solve it. The solution is surprisingly simple, yet I still believe this puzzle deserves a difficulty rating of 9/10. Annoyingly when I first came across it there were no hints, and I could only solve it after hearing the first stage of the solution which is a very big hint.

There are a countably infinite number of logicians who are all be given a hat with a real number number on it. The logicians cannot see their own number (including via reflective surfaces or other mechanisms) and they are not allowed to communicate with others. Simultaneously they will announce what they believe to be the number on their hat. They win the game if at most finitely many of the logicians guess correctly. They are allowed to strategise beforehand - what strategy should they use?

- There is no pattern or connection between these numbers - they are selected completely independently from each other and no information can be gained about what a given number is using knowledge of other numbers.
- The logicians can see the number of every other logician.
- All the logicians can be trusted to always follow the strategy perfectly (although if there were finitely many bad actors who deliberately guessed wrong this would not make a difference).
- The logicians are capable of performing any mathematical operation and have no bound on how much information they can remember.
- You may assume the axiom of choice.

Hints:

1. The logicians will not know whether they guessed correctly, even after everyone has announced their guess. This information is impossible to determine because of the independence of the numbers. Do not focus on a single logician - only the collective matters.
2. The logicians can decide a priori on an ordering of themselves, and this gives a sequence of real numbers. Consider the space of sequences of real numbers.
3. Construct a suitable equivalence relation between sequences of real numbers.
4. Define two sequences to be “close” if they differ in only finitely many positions.

Solution:

As there are a countable number of logicians there exists an ordering of them which they can agree on in advance. Taking each logicians number in this order defines a sequence. Consider two sequences (a_n) and (b_n) and define $I_{a,b} = \{n \mid a_n \neq b_n\}$ to be the indices where (a_n) and (b_n) differ. We say (a_n) (b_n) if and only if $|I_{a,b}|$ is finite. This equivalence relation captures the idea that two sequences are close. First we show this is in fact an equivalence relation.

The relation is clearly reflexive as a sequence differs from itself in zero locations which is finite. (a_n) differs from (b_n) in exactly the same places that (b_n) differs from (a_n) and thus symmetry is inherited from the non-equals relation. For sequences (a_n) , (b_n) , and (c_n) , we see the relation is transitive by equation 1.

$$|I_{a,c}| \leq |I_{a,b} \cup I_{b,c}| \leq |I_{a,b}| + |I_{b,c}| \leq \infty \quad (1)$$

This equivalence relation partitions the set of real sequences into equivalence classes. Within each equivalence class, each sequence is close to each other sequence. Each sequence is in an equivalence class, and no sequence is in more than one equivalence class. The first follows as a sequence must be close to itself, and the second follows by transitivity. Using the axiom of choice, an element of each equivalence class can be chosen. This is done for each equivalence class, and the logicians can all agree which sequence is chosen for each equivalence class.

Let the sequence defined by the numbers on the logician's hats be (x_n) . Logician k can define a sequence (y_n^k) by $x_1, \dots, x_{k-1}, y, x_{k+1}, \dots$ where y is the unknown number that is on their hat. For all possible values of y , (y_n^k) will belong to the same equivalence class as (x_n) as they only differ in position k . Each logician can do this and they will each identify the same equivalence class by transitivity (This follows because $(y_n^{k_1}) (y_n^{k_2}) \forall k_1, k_2 \in \mathbb{N}$). All logicians agree on the representative sequence, (z_n) for that equivalence class.

The strategy for the k^{th} logician is to guess z_k . The resulting sequence of guesses must differ in only finitely many locations as $(z_n) (x_n)$.

Extensions and Comments:

I think this puzzle seems so hard because it is natural to focus on how a single logician could get the answer correct. Suppose there was only one logician with a natural number on their head, and they had as many guesses as they like to guess the number. We would expect them to need an infinite number of guesses before they guessed correctly, and as the natural numbers are countable this seems far easier than guessing a number from an uncountable set such as the real numbers.

Having only a finite number guess wrong seems very restrictive given how much smaller any finite number is than infinity. When seeing the answer however I thought that the number being finite seemed less restrictive. Once the equivalence relation puts us in the domain of dealing with finite numbers, all over control follows by pushing any difficulties under the rug of finite + finite = finite. This problem highlighted to me how hard it is to have a good intuition for finite numbers vs infinity.

I believe we can even construct a more restrictive version of this problem. Suppose all the logicians were in a line facing the same direction such that they can see all the hats in front of them but none of the hats behind them. The logician in position k would not be able to see $k - 1$ numbers, but given that this is a finite number they would still be able to identify which equivalence class the sequence lived in. The strategy should work exactly the same way. As noted in the hints, any finite number of malicious actors who do not conform to the strategy can also be tolerated.

Part of my frustration with this problem was that I could not get a foothold into it, although once the equivalence relation on real sequences was given I saw the rest of the solution immediately. To me this highlights the importance of hints in a problem, and in particular giving hints in a sequence where as little information is imparted at a time. I do not think I would have got this without hints, but I feel the hint I did receive robbed me of solving the problem properly.

1.9 The Chessboard Encoding Problem

This problem is from a collaboration between 3blue1brown and Matt Parker. I attempted this problem years ago, but could not do it, but on a second attempt recently I got it instantaneously. I think I spoiled this problem for myself however as I had watched a video recently about a highly relevant problem, and I would never have got it so fast otherwise. I give this problem a difficulty of 5/10, although the ideas behind the solution did lead to someone winning the Turing Prize.

You and a colleague are imprisoned, although the prison warden gives you an opportunity for freedom. There is an 8×8 chessboard where each square is a hidden compartment, and the warden reveals to one of you, let us call them person A, which square the key to the door is hidden in. A coin is then placed on each square in a configuration decided by the warden that is not known to you or your colleague in advance. Person A has to turn over exactly one coin before leaving the room, at which point person B enters and needs to determine which square has the key. You are allowed to agree a strategy beforehand, although the warden will hear your strategy, and will try and find a configuration of coins to make your strategy not work.

Hints:

1. Consider a 1×2 board, then a 2×2 board (this problem has a solution if and only if the number of squares is a power of 2, see the extensions and comments for a proof).
2. For the 2×2 board, if you can encode both the column and the row then you have encoded the entire position. Think of those two bits of information like binary bits.
3. Consider the parity of the number of coins that are heads-up in a region.

Solution:

Suppose we had a 1×2 chessboard. We could encode whether the coin is on the left or the right by considering whether the coin on the left is heads-up. If the coin on the left is already a heads and we need to encode that the key is on the left then we can freely flip the coin on the right.

Considering a 2×2 chessboard is the first non-trivial case. We will try and extend the idea from the 1×2 chessboard to encode whether the key is on the left side or the right side. Using the top left coin as an indicator for this would work, but we may not be able to also encode whether the key is on the top or bottom half. If we wanted to encode the column with a similar strategy, let us say for example by using the top right coin as an indicator, we may need to flip two different coins.

The solution is to use two squares as an indicator for both, and have these two squares overlap. Before we look at how we could use two multiple squares as an indicator for one bit of information, we demonstrate how they could be used. Here there would be four squares to flip:

1. The overlap between the row indicator pair and the column indicator pair
2. In the row indicator pair, but not in the column indicator pair
3. In the column indicator pair, but not in the row indicator pair
4. Not in the row indicator pair or the column indicator pair

Before flipping any coins, the existing state of the board will indicator some row and column pair. If this is the correct row and column then we are done, and we would flip the coin on square 4 which does not change either of our indicators. If it is in the wrong row but the correct column, then we flip the coin on square 2. The other cases follow similarly.

We can use multiple squares as an indicator by considering the parity of the number of heads. If there are an even number of heads, we say it encodes 0, or false, and if there are an odd number of heads we

say it encodes 1, or true. This can be done for regions of any size. The key to our solution for a 2×2 board work was that the parity of each region could be controlled independently.

The problem now is how to construct such regions for a larger board. If we flatten the 2×2 board into a 1×4 board by taking the bottom row and appending it on the right-hand side then we can map the strategy from before onto this new board. Being in the left column now means being in either the first or third square and being in the top row now means being in either the first or second square.

For an 1×8 board we can extend this again, having the first four squares encoding left vs right side, the first and third disjoint pairs encoding the disjoint pair, and the odd-numbered squares encoding the oddness or the square. The way we extended the 1×2 board to a 2×2 board can now be done to get an 8×8 board. There are many ways of creating these regions however and we can modify our region choice slightly to give a nicer answer.

As alluded to before, we want to produce a six-digit binary number which can be used to encode a square of the chessboard as $8 \times 8 = 64 = 2^6$. The six regions shown in figure 1 allow this.

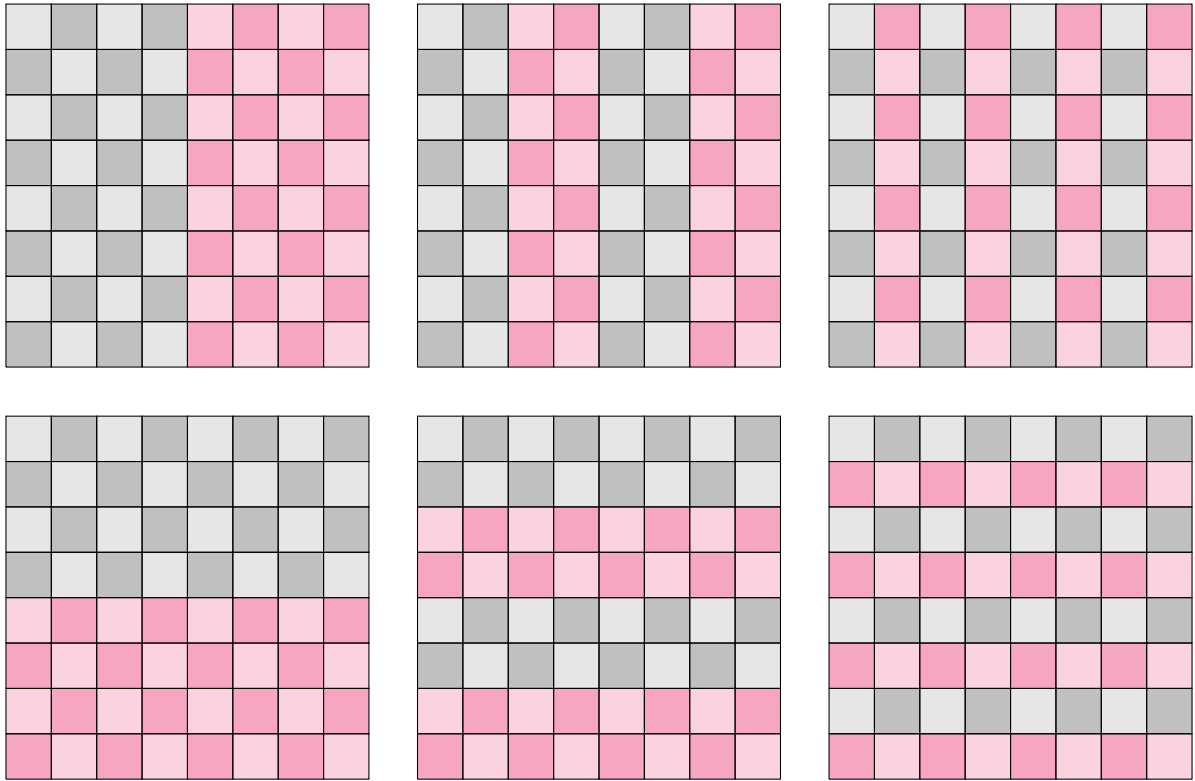


Figure 1: Six regions that each define a binary bit via the parity of the number of heads-up coins in the squares highlighted in red.

In the example shown in figure 2, the key has been hidden in the cell in the third row and seventh column. Using the first three digits to encode the column number and the last three digits to encode the row number, this is encoded as 111011. In the red highlighted region in figure ?? there are 19 heads which is equal to 1 modulo 2. This matches the first digit in the number we want to encode which means we want to preserve the parity of heads in this region. Therefore we will flip a coin outside the red highlighted region, that is, on the left hand side. We can do a similar process for the other regions and the results of this are given in table I

T	T	T	H	H	T	T	H
T	T	H	H	H	H	H	H
T	T	H	H	H	H	H	T
T	T	T	H	T	T	H	H
H	T	H	H	T	H	H	H
H	T	H	T	T	H	H	T
T	H	T	T	T	T	T	H
H	T	T	H	T	H	H	T

Figure 2: An example of random coins on a chessboard. The red highlighted square indicates the position of the key.

Table I: An example of how to determine which regions to flip the parity of

Region	Head Count	Head Parity	Encoded Parity	Parity Change	Flip Red Region
1	19	1	1	0	No
2	21	1	1	0	No
3	17	1	0	1	Yes
4	16	0	0	0	No
5	14	0	1	1	Yes
6	17	1	0	1	Yes

Reading from the last column, we need to flip the coin with encoding 001011 which is the the second column and third row. Once this is done, the parities of the different regions would be read off as 111011, precisely giving the position of the square with the hidden key. Note that the position of the coin to flip is given by the XOR of the state encoded by the position of the hidden key and the state encoded by the board.

Extensions and Comments:

We notice that as there are 2^6 different states to encode and 2^{64} boards, there must be 2^6 sets of 2^{58} elements that are equivalent in some sense. We can make this intuition more rigorous. Consider the set of 64-tuples of binary numbers, $G = \{0, 1\}^{64}$. For $x, y \in G$ we can define the binary option, $x \circ y := (x + y) \bmod 2$, and note that this is equivalent to elementwise XOR. This can easily be seen to form an abelian group. Elements of this group represent states of the board. We can also make a group where each element corresponds to a square on the board.

Define $f(x)$ to be the function that takes $x \in G$ to the index of the cell that it encodes. With an abuse of notation using $\sum$ to mean repeated application of $\circ$, we can give this function explicitly as $f(x) = \sum_{i \in I_k} x_i$ where I_k is the indexing set corresponding to the k^{th} digit in the encoding. If we define $K := \ker f$ then it follows from the first isomorphism theorem that K is a normal subgroup of G and the quotient group $H = G/K$ is well-defined. K represents all the board states that encode 0 and we are dividing out this information. This means that we expect each element of H to represent all the board states that encodes the same square.

We could arrive at the same group through a congruence. Define an equivalence relation between board states where two states are equivalent if they encode the same cell, that is, $x \sim y \iff f(x) = f(y)$. It is easy to show this is an equivalence as the equivalency axioms are inherited from the equality relation. Equation (2a) demonstrates that mapped elements are self-inverse from which it follows in equation (2b) that the congruence relation respects inverses. Equation (2c) shows that f is linear and thus the congruence respects the binary operation as well, as shown in (2d). This is essentially equivalent to saying that flipping the parity of a coin within a region flips the parity of the whole region.

$$f(x) \circ f(x) = 0 \implies f(x) = f(x)^{-1} \quad (2a)$$

$$f(x) = f(y) \implies f(x)^{-1} = f(y)^{-1} \quad (2b)$$

$$f(x)_k \circ f(y)_k = \sum_{i \in I_k} x_i \circ \sum_{i \in I_k} y_i = \sum_{i \in I_k} x_i \circ y_i = \sum_{i \in I_k} (x \circ y)_i = f(x \circ y)_k \quad (2c)$$

$$\begin{aligned} x^1 \sim y^1, x^2 \sim y^2 &\implies f(x^1)_k = f(y^1)_k, f(x^2)_k = f(y^2)_k \quad \forall k \\ &\implies f(x^1 \circ x^2)_k = f(x^1)_k \circ f(x^2)_k = f(y^1)_k \circ f(y^2)_k = f(y^1 \circ y^2)_k \quad \forall k \\ &\implies x^1 \circ x^2 \sim y^1 \circ y^2 \end{aligned} \quad (2d)$$

1.10 Pills While Blindfolded

This is a nice and accessible problem that requires some out of the box thinking. Good problem for when walking as paper would not help much. I give this problem a 1/10 in difficulty.

An evil logician has poisoned you, but gives you a way to survive. They offer you two red pills and two blue pills and to survive you need to take exactly one red and one blue pill. The only complication is that you are blindfolded. What is your strategy to survive? Standard evil logician rules apply.

Hints:

1. There is no way to gain any information about what colour the pills are. This means your strategy would have to work for any permutation of the pills.
2. Taking all the pills would give you exactly double the dose.
3. Consider the same problem but with one red pill and one blue pill and needing to take half of each. Apply this to the problem with four pills.

Solution:

Arrange all the pills in a line, such as “O O O O”. Instead of splitting them vertically down the middle like “O O | O O”, cut them horizontally. Each half will contain exactly two lots of half a red pill and two lots of half a blue pill. This is the same amount as one red pill and one blue pill.

Extensions and Comments:

It is not stated whether you can or cannot cut the pills in the problem's statement, but I think if you gave this information it would completely give away the solution. I think it is better to leave this as the creative option. This problem works with any number of pills and any number of colour of pills.

1.11 25 Horses

This is a nice simple problem where a pen and paper is not necessary so you can give it to someone on a walk. I rate this problem 2/10 in difficulty.

There are 25 horses of unknown speed. They can be raced in groups of five to find the finishing order, but their times are not recorded. How many races are necessary to determine the fastest three horses and what strategy should be used to accomplish this?

Hints:

1. Can you think of an easier sub-problem involving the 25 horses?
2. How would you determine the fastest horse?
3. The second-fastest horse may have been in the same race as the fastest horse. Keep track of where the possible fast horses may be and then race them together.

Solution:

All horses need to be raced at some point and 25 naturally decomposes into five races of five horses which we will call the heats. The fastest horse must be among the winners of the heats and racing these five together in a sixth race determines the winner. We label the heats A, B, C, D, and E in order of how fast the fastest horse in each race was, and the horses in these races by their position. For example, B3 is the horse that came third in the heat with the horse that came second in the sixth race.

The second fastest horse would have either won their heat or have been beaten by the fastest horse. This means they were either A2 or B1. We cannot distinguish between these cases with the information we have so a seventh race is necessary. The third fastest horse could be any of the following three:

1. A3: This is where the second fastest horse was in the same heat as the fastest horse and both of them beat the third fastest.
2. B2: This is where the second fastest horse was in a different heat from the fastest horse, but shared a heat with the third fastest.
3. C1: The top three fastest horses were in different heats so all made it through to the sixth race.

Together this makes A2, B1, A3, B1, and C1 which is five horses. If these horses are raced together then the top two positions give the second and third fastest horses.

Extensions and Comments:

The different possibilities for the final race are given in table II.

Table II: Final race possibilities based on position of the fastest three horses

Race			Horse Order	
1	2	3	Second	Third
1, 2, 3			A2	A3
1, 2	3		A2	B1
1	1, 2		B1	B2
1, 3	1		B1	A2
1	2	3	B1	C1

2 NUMBERS

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2.1 Six Digit And Two Set Comparison

I found this problem in a book of problems from the Grade Five Leningrad Mathematical Olympiad [1], in particular, this is question two from 1989. I have only found one person who could solve this as of 2024 so it may be trickier than it seems given it is designed for 11 year old children.

Consider the set of six digit numbers including leading zeros, that is, 000000 to 999999. Set X is defined as all such numbers where the first three digits sum to the same amount as the sum of the last three digits. Set Y is defined as all such numbers where all the digits add up to 27. Show that sets X and Y are the same size.

Hints:

1. The size of each set does not need to be determined.
2. The fact that $27 = 3 \cdot 9 = \frac{6}{2} \cdot (\text{base} - 1)$ is not a coincidence.
3. Make a bijection between the sets.

Solution:

Denote the number as the concatenation of six digits, $abcABC$, and using the same notation consider the map $\phi(abcABC) = (9 - a)(9 - b)(9 - c)ABC$. Intuitively it can be seen that ϕ uniquely pairs up numbers between the two sets, meaning that they must be the same size as otherwise one set would not be big enough to produce pairs for the other set.

To prove this, it is shown that ϕ is a bijective involution between the sets, implying that they are the same size. First we see that ϕ is well-defined, as each digit from 0 to 9 is mapped to another digit from 0 to 9. Therefore the result of applying ϕ to a six digit base ten number is another six digit base ten number, and each digit in the input only affects the digit in the same position in the output. The following implications show that ϕ maps X to Y .

$$\begin{aligned}
 abcABC &\in X \\
 \iff a + b + c &= A + B + C \\
 \iff 27 &= (9 - a) + (9 - b) + (9 - c) + A + B + C \\
 \iff 27 &= \text{Digit sum}(\phi(abcABC)) \\
 \iff \phi(abcABC) &\in Y
 \end{aligned}$$

It is clear that $\phi(\phi(abcABC)) = abcABC$. This implies that it is impossible for ϕ to map two distinct inputs to the same output, as otherwise ϕ would need to map that result to both inputs simultaneously. This also implies that every six digit number must be in the image of ϕ , meaning the reverse implications presented previously combined with ϕ^2 being the identity map implies that ϕ maps Y to X . Together this shows that X and Y are at least as large as each other, and therefore the same size.

Extensions and Comments:

A similar problem can be given in any base, and for any even number of digits with a near identical proof. I like this problem because it uses a way of understanding size that is very fundamental, but people do not usually think about. The size of each set does not need to be determined, but it can be found by taking square of the number of occurrences for each unique digit sum, and adding together. This gives 55252 as the size of X and Y .

The relationship between the sums of the first and last three digits and the total sum is actually much closer than this problem suggests. In the notation used previously, define the difference to be $A + B + C - a - b - c$, and the total to be the digit sum. Define X_d to be the set of numbers that have the same difference, and define Y_t to be the set of numbers that have the same total. Almost identical

reasoning used to solve this puzzle can be used to show that $|X_{d-27}| = |Y_t|$. Additionally, a map that subtracts each digit from 9 can be used to show that $|X_d| = |X_{-d}|$ and $|Y_t| = |Y_{54-t}|$. This property extends to other bases and other even number of digits, and is demonstrated explicitly in table III.

Table III: The distribution of the size of X_d and Y_t

Difference		Total	
Value	Count	Value	Count
-27	1	0	1
-26	6	1	6
-25	21	2	21
-24	56	3	56
-23	126	4	126
-22	252	5	252
-21	462	6	462
-20	792	7	792
-19	1287	8	1287
-18	2002	9	2002
-17	2997	10	2997
-16	4332	11	4332
-15	6062	12	6062
-14	8232	13	8232
-13	10872	14	10872
-12	13992	15	13992
-11	17577	16	17577
-10	21582	17	21582
-9	25927	18	25927
-8	30492	19	30492
-7	35127	20	35127
-6	39662	21	39662
-5	43917	22	43917
-4	47712	23	47712
-3	50877	24	50877
-2	53262	25	53262
-1	54747	26	54747
0	55252	27	55252
1	54747	28	54747
2	53262	29	53262
3	50877	30	50877
4	47712	31	47712
5	43917	32	43917
6	39662	33	39662
7	35127	34	35127
8	30492	35	30492
9	25927	36	25927
10	21582	37	21582
11	17577	38	17577
12	13992	39	13992
13	10872	40	10872
14	8232	41	8232
15	6062	42	6062

2.2 Appending Cyclicly Mapped Sequence

This problem was part of the coding assessment for Blackrock which I was helping a friend with. I originally did this in the naive way as I had the power of Python, but looking at it further I realised there was a simple solution. This puzzle does not require any large leaps in understanding and the solution builds quite naturally. I consider this problem a 4/10 in terms of difficulty.

At the 0th iteration, a sequence is the single digit 0. On each iteration, all zeros are mapped to one, all ones are mapped to two, and all twos are mapped to zero. The transformed sequence is then appended on to the end of the original sequence to get the sequence for the next iteration. The question is to find a nice method or expression to determine the value of the digit at index n where the first digit is defined to be at index 0. The first few terms are shown below.

```

0
01
0112
01121220
0112122012202002

```

Hints:

1. It is not necessary to generate the whole sequence.
2. Initially forget about taking the result modulo 3, and assume a base with infinitely many symbols.
3. Convert n to binary.

Solution:

The digit sum of the binary digits of n modulo 3 gives the answer. At any given iteration, an index is either part of the original sequence in the previous iteration, or is part of the appended sequence. The digit would have had one added to it modulo three at an iteration if and only if it was in the appended part at that iteration, and therefore the history of each position should be understood.

Take 13 as an example. At the fourth iteration, it is in the appended part as it is larger than 8. The mapped digit was $13 - 8 = 5$. At the third iteration this index was also in the appended part as it is larger than 4, and the mapped digit was 1. This was in the original part of the sequence at the second iteration, and in the appended part at the first iteration. Going from the last iteration backwards gives a history of appended, appended, original, appended, matching the binary expansion, 1101.

To see this relationship with the binary more generally, note that at iteration n , the index k is in the appended part if and only if the $k \bmod 2^{n-1}$ index is in the appended part. This is the case if and only if the n^{th} binary digit of k is 1, as the binary expansion of $k \bmod 2^{n-1}$ is the same as that of k but truncated to leave the last n digits. To get the digit at the k^{th} position, it is only necessary to find the number of times it was in the appended part as this is the number of times a 1 was added to it. This corresponds exactly to the sum of the digits in the binary expansion of k , and mod commutes with addition, therefore the modulus can be taken after the sum.

Extensions and Comments:

This solution builds quite naturally and an initial approach of testing some examples does well to give an intuition as to how the problem works. I think this would work better as a supervised interview question instead of as a coding test as there is nice problem solving here. The only line of Python code needed is `sum([int(i) for i in bin(n)[2:]]) % 3`. This problem easily extends for transformation rules that use addition of any constant modulo any other positive natural number by the same argument, although with the extra step of multiplying the sum of binary digits by the constant added to the appended part. The time and memory complexity of the solution offered here is $O(\log n)$, whereas the naive method of generating the whole sequence needed and selecting the digit has a complexity of $O(n)$.

2.3 5 by 6 Magic Rectangle

This is question 1 from the 1979 Grade 5 Leningrad Mathematical Olympiad [1], so it was designed for children aged around 11. Despite this, most people I ask have not been able to solve it and give up very quickly. Personally I think the success rate would be higher if people were more tenacious, and I give this problem a difficulty of 2/10.

The numbers 1 to 30 are each to be placed in a 5 by 6 grid without repeats such that all the rows sum to the same total, and all the columns sum to the same total. Determine whether this is possible, and if so, provide a solution.

Hints:

1. Be more systematic than trying to move the numbers around to get the same totals.
2. Think about how you would calculate the totals for the rows and columns.

Solution:

This problem becomes easy when you think about it systematically. Before placing any numbers in the table, the totals for the rows and columns should be found as a target to aim for. The sum of all the column totals will be equal to the sum of all the numbers in the grid, and similarly for the rows. This total is an invariant of the problem, any arrangement of numbers will give the same total sum. This invariant can be determined using the formula for the sum of the first n natural numbers, giving $S = \frac{1}{2} \cdot 30 \cdot 31 = 15 \cdot 31$. As the column total is the same for all six columns, the sum of the column totals will be a multiple of six. This means the sum of all numbers in the grid is also a multiple of 6. $S = 15 \cdot 31$, which are both odd, therefore the column total cannot be a whole number. As the column total is the sum of whole numbers, this is a contradiction and the task is impossible.

Extensions and Comments:

This method can show that such a grid is impossible if the dimensions of the grid are of different parities. This can be done by substituting a general odd number and a general even number, $n = 2a$ and $m = 2b + 1$ respectively, into the formula $S = \frac{1}{2} \cdot nm \cdot (nm + 1)$. This is done in equation (3) and it is seen to be an odd multiple of a as $\text{ODD} \cdot \text{ODD} = \text{ODD}$. For S to divide n it needs to be an even multiple of a and thus we get a contradiction as before.

$$S = \frac{1}{2} \cdot 2a \cdot (2b + 1) \cdot (2a \cdot (2b + 1) + 1) = a \cdot (2b + 1) \cdot (2(ab + a) + 1) \quad (3)$$

In the case of two even numbers or two odd numbers, the corresponding equation for S divides both n and m . This is seen in equations (4a) and (4b) respectively. Passing this test does not imply that such a magic rectangle is possible however, for example there are only two possible 2 by 2 grids and neither of them are magic.

$$\begin{aligned} S &= \frac{1}{2} \cdot 2a \cdot 2b \cdot (2a \cdot 2b + 1) \\ &= n \cdot b \cdot (2a \cdot 2b + 1) \\ &= m \cdot a \cdot (2a \cdot 2b + 1) \end{aligned} \quad (4a)$$

$$\begin{aligned} S &= \frac{1}{2} (2a + 1) \cdot (2b + 1) \cdot ((2a + 1) \cdot (2b + 1) + 1) \\ &= (2a + 1) \cdot (2b + 1) \cdot (2ab + a + b + 1) \\ &= n \cdot m \cdot (2ab + a + b + 1) \end{aligned} \quad (4b)$$

2.4 Irrational to High Power Near Integer

This problem was given to me by my Oxford admissions preparation tutor, Jeremy. The original problem he gave was “what happens, now prove it”, where I found the first part, and which I have decided to reveal upfront. I came back to this problem in 2024 and solved it in around 20 minutes and I have provided this proof as the second solution. Dr Barker on YouTube also covered this problem and gave the much slicker argument which is given as the first solution. This problem benefits greatly from pen and paper, and is much more suited to the mathematically inclined. I rate this problem a 5 out of 10 in terms of difficulty.

$(1 + \sqrt{2})^n$ gets close to an integer when raised to a high power. Explain why this is the case. Extension question: characterise exactly which $a + \sqrt{b}$ where this occurs.

Hints for Solution 1:

1. Use the binomial expansion to expand $(1 + \sqrt{2})^n$. How would you need to modify this to make the odd powers of $\sqrt{2}$ disappear?
2. Consider the conjugate, $(1 - \sqrt{2})^n$.
3. Add the conjugate to the original to make the odd powers cancel out.

Solution 1:

We define $a_n = (1 + \sqrt{2})^n$ and b_n as the sum of $(1 + \sqrt{2})^n$ and its conjugate. Using the binomial expansion we see in equation (5) that the odd powers of $\sqrt{2}$ cancel out, and thus b_n is an integer. We also note that $|1 - \sqrt{2}| < 1$, and therefore it annihilates when raising it to increasingly higher powers. This means $|a_n - b_n| \rightarrow 0$ as $n \rightarrow \infty$, and thus a_n converges to an integer.

$$\begin{aligned}
 b_n &= (1 + \sqrt{2})^n + (1 - \sqrt{2})^n \\
 &= \sum_{k=0}^n \binom{n}{k} (1^{n-k} \cdot \sqrt{2}^k + 1^{n-k} \cdot (-\sqrt{2})^k) \\
 &= \sum_{k=0}^n \binom{n}{k} \sqrt{2}^k \cdot (1 + (-1)^k) \\
 &= \sum_{k=0}^n \binom{n}{k} \sqrt{2}^k \cdot \{k \text{ is even}\} \\
 &= \sum_{m=0}^{\lfloor \frac{n}{2} \rfloor} \binom{n}{2m} 2^m \in \mathbb{N}
 \end{aligned} \tag{5}$$

Hints for Solution 2:

- 1.

Solution 2:

Extensions and Comments:

2.5 Spherical Polar Basis

Hints:

- 1.

Solution:

Extensions and Comments:

2.6 Interpolating Polynomials

This subsection is about a series of results I found about interpolating polynomials that are all linked together. I found these at several different points throughout my mathematical journey, and have attempted to present them in a way that highlights how one might discover them for themselves.

2.6.1 Calculus of Differences and n^{th} Term Rule

The n^{th} term rule is a function from the natural numbers excluding 0 to the reals that generates a sequence. In year 9 we were shown a method to find the n^{th} term rule for a sequence generated by a quadratic polynomial that confused me, and when I investigated this for myself I found an extension to higher orders. I did not prove that this always worked at the time, and did so when I revisited the topic during my maths masters. Determining a cubic polynomial for a four term sequence was one of the very first programs I wrote soon after I turned 15. During the winter holiday I wrote a version that could find the polynomial for arbitrary length sequences.

2.6.2 The Lagrange Interpolating Polynomial

In my second year of undergrad I was doing problem sheet 1 from part A linear algebra [2], and question two was the following.

(Harder) Show that the space of functions $f : \mathbb{N} \rightarrow \mathbb{R}$ does not have a countable basis.

I was not initially convinced by this and thought that a polynomial interpolating all the points of f would be a counterexample. I have discussed why this reasoning is wrong in the extensions and comments, along with the answer to the original question. I am also not sure why I did not realise I had my supposed counterexample from the calculus of differences method, but nonetheless I found a new method that is the subject of this exercise. If I was a teacher then I would definitely be giving this problem to my students.

Given a finite set, $A \subset \mathbb{R}$, and a function $f : A \rightarrow \mathbb{R}$, find an interpolating polynomial of lowest degree, $p : A \rightarrow \mathbb{R}$. That is $f(x_i) = p(x_i) \quad \forall x_i \in A$. This is known as the Lagrange Interpolating Polynomial, or simply the Lagrange Polynomial. Finding an explicit formula is the first part of the question, and showing it is minimal is the second part.

Hints:

1. Use linearity - the finite sum of a collection of polynomials is also a polynomial.
2. Find a way to focus on one point at a time.
3. Use the factor theorem - a polynomial has a root at x_0 if and only if it divides $x - x_0$.

Solution:

The key to this problem is to use the fact that two polynomials can be added to get another polynomial. For each point, (x_i, y_i) , we can find a polynomial, $p_i(x)$, that is 0 when evaluated on all x_j where $j \neq i$ and satisfies $p_i(x_i) = y_i$. The sum of these polynomials gives us an interpolating polynomial by equation (6).

$$p(x_i) = \sum_j^n p_j(x_j) = \sum_j^n y_j \delta_{ij} = y_i \quad (6)$$

The problem now reduces to finding each of the p_i . In fact we can simplify the problem further as we only need to find a function, q_i , that is 0 for x_j for $j \neq i$ and satisfies $q_i(x_i) \neq 0$. Once such a function has been found, we can define p_i as in equation (7a). We can now deduce q_i using the factor theorem:

x_0 is a root of p if and only if $p(x)$ divides $x - x_0$. This gives q_i as the product in equation (7b). The final polynomial is given in (7c).

$$p_i(x) = \frac{q_i(x)}{q_i(x_i)} \cdot y_i \quad (7a)$$

$$q_i(x) = \prod_{\substack{j=1 \\ j \neq i}}^n (x - x_j) \quad (7b)$$

$$p(x) = \sum_{i=1}^n \prod_{\substack{j=1 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j} y_i \quad (7c)$$

Now that an interpolating polynomial has been constructed, it is necessary to show it is of minimal order. We assume that the n coordinates, (x_i, y_i) , are interpolated by a polynomial of order one less than what is needed in general, $\sum_{k=0}^{n-2} a_k x^k$. This gives a formula for y_i in terms of x_i which can be substituted into equation (7c) and defines the polynomial $\sum_{k=0}^{n-1} b_k x^k$ as demonstrated in equation (8).

$$p(x) = \sum_{i=1}^n \sum_{j \neq i} \frac{x - x_j}{x_i - x_j} y_i \quad (8a)$$

$$= \sum_{i=1}^n \sum_{j \neq i} \left(\frac{x - x_j}{x_i - x_j} \sum_{k=0}^{n-2} a_k x^k \right) \quad (8b)$$

$$= \sum_{k=0}^{n-2} a_k \sum_{i=1}^n \sum_{j \neq i} \frac{1}{x_i - x_j} (x^{k+1} - x_j x^k) \quad (8c)$$

$$= \sum_{k=0}^{n-1} b_k x^k \quad (8d)$$

The aim is to show that the generated interpolating polynomial has a degree no higher than necessary to interpolate the points. This means we need to show that $b_{n-1} = 0$ which is done below in equation (9). In equation (9c) we have made the substitution $i = n+1-l$ and $j = n+1-m$. It is easy to see that the range $i = 1, \dots, n$ is equivalent to $l = 1, \dots, n$, and the same for j and m respectively (the other is just reversed) so the outer sums are the same. $i < j \iff n+1-l < n+1-m \iff -l < -m \iff l > m$

shows that the inner sums are over the same range as well.

$$b_{n-1} = a_{n-2} \sum_{i=1}^n \sum_{j \neq n} \frac{1}{x_i - x_j} \quad (9a)$$

$$= a_{n-2} \left(\sum_{i=1}^n \sum_{j < i} \frac{1}{x_i - x_j} + \sum_{i=1}^n \sum_{j > i} \frac{1}{x_i - x_j} \right) \quad (9b)$$

$$= a_{n-2} \left(\sum_{i=1}^n \sum_{j < i} \frac{1}{x_i - x_j} + \sum_{l=1}^n \sum_{m < l} \frac{1}{x_l - x_m} \right) \quad (9c)$$

$$= a_{n-2} \left(\sum_{i=1}^n \sum_{j < i} \frac{1}{x_i - x_j} + \sum_{i=1}^n \sum_{j < i} \frac{1}{x_i - x_j} \right) \quad (9d)$$

$$= a_{n-2} \sum_{i=1}^n \sum_{j < i} \left(\frac{1}{x_i - x_j} - \frac{1}{x_i - x_j} \right) \quad (9e)$$

$$= 0 \quad (9f)$$

So far we have only demonstrated that the generated interpolating polynomial will be of minimal order when the n coordinates to be interpolated can be fit by a polynomial of degree $n - 1$ or $n - 2$. It is therefore necessary to show that even more degenerate sets of coordinates also give rise to minimal order interpolating polynomials. Suppose the coordinates can be interpolated by a polynomial of degree $k - 1$ where $k < n$.

Choosing any k of the n coordinates will generate a minimum degree interpolating polynomial. We will show later that the minimum degree polynomial is unique, and therefore the same polynomial is generated by any of the k coordinates. By the above algebra we know that interpolating any $k + 1$ coordinates will generate the same polynomial as generated by a subset of the $k + 1$ coordinates of size k . This logic can be continued by induction, eventually interpolating all n coordinates.

Extensions and Comments:

2.6.3 Properties of Resulting Polynomials

3 PROBABILITY

3.1	Sum of Many Dice	37
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3.1 Sum of Many Dice

I thought of this puzzle while was a trainee teacher. It is reasonably simple but employs a nice trick. Everyone I have asked this as of the time of writing has not got it, but my dad went straight for the solution and said that was just the obvious way of solving it. I am inclined to agree, but given how most people have not got it, I rate this puzzle a 3/10 in terms of difficulty.

Three regular six sided dice are rolled and the total is found. what is more likely to happen, the sum is equal to 9, or 10?

Hints:

1. It is not necessary to enumerate all combinations that give a total of 9 and 10.
2. Think about the shape of the distribution of the total of the dice rolls.
3. Calculate the mean of the distribution.

Solution:

The mean of one dice is 3.5, and by linearity of expectation the mean of the sum of three dice is 10.5. The probability is lowest for the sums of 3 and 18 which can only be achieved in one way, and increase as they get to the middle with the peak at 10 and 11. The exact probability does not need to be computed as only the order is of interest, and this can be determined by the distance from the peak. 9 is further from 10.5 than 10 is, therefore it has a lower probability.

To give a better intuition as to why the probability is increasing up to the middle and then decreases, consider the state space as a 6 by 6 by 6 cubic grid. Each of the dimensions represents one of the dice, and each cell represents one possible result of the three dice. Cells with the same total lie on the same diagonal plane ($x + y + z = p$ if the axes of the cube lie on the coordinate axes), and each cell has the same probability of occurring. This means the highest probability total is where the plane has the largest intersection with the cube. As the plane moves through the cube, the intersection can be seen to increase until it is halfway through, after which it decreases.

Extensions and Comments:

I like this problem because it is very simple and the solution easily cuts away what is unnecessary. This problem can very easily be extended for any number of dice as long as there are more than one dice², and any number of sides on the dice. This even works if the dice are different sizes, the distribution will still be convex by the same argument, and the mean can be found by adding the individual means of the dice. I imagine if a version of this problem was given with a hundred dice and totals of 364 and 335 then people would be more suspicious and get the answer faster.

²If there is only one dice then the distribution is not strictly convex from below, therefore the probabilities are weak inequalities rather than strict.

3.2 The Umbrella Problem

Hints:

- 1.

Solution:

Extensions and Comments:

3.3 The Cupboard Game

I found this problem on Reddit and like many good probability problems, the usual intuition is wrong. In hindsight I should have verified my intuition by explicit calculation, but I was lazy and did not bother so got it wrong. I rate this a 4/10 in difficulty, although I doubt anyone would get the answer to this very fast.

There are eight cupboards arranged in two rows of four columns as shown below. Inside a randomly chosen cupboard there is a prize and Alice and Bob play a game to determine who gets the prize. Alice opens cupboards going along the rows first, so in order A, B, C, D, E, F, G, H . Bob opens cupboards going along the columns first, so in order A, E, B, F, C, G, D, H . Both open cupboard doors at the same time and if the prize is inside a cupboard they have both picked simultaneously then it is a tie. Is one of them more likely to win, and if so, who?

A	B	C	D
E	F	G	H

Hints:

1.

Solution:

Turn	Player		Prize Location							
	Alice	Bob	A	B	C	D	E	F	G	H
1	A	A	Draw							
2	B	E		Alice			Bob			
3	C	B			Alice					
4	D	F				Alice		Bob		
5	E	C								
6	F	G							Bob	
7	G	D								
8	H	H								Draw

Extensions and Comments:

3.4 The Two Child Problem

This is a classic problem I believe originally posed by Martin Gardner, although I have given it here with the unambiguous phrasing. I do not know where the harder variant comes from. I rate this problem a 2/10 in difficulty.

Easy: If a family with two children, one of which is a girl, is chosen at random, what is the probability that the other child is also a girl? **Harder:** If a family with two children, one which is a girl born on a Tuesday, is chosen at random, what is the probability that the other child is a girl?

Hints:

1. Consider the state space.
2. Each item in the state space is equally likely.
3. Use Bayes' theorem.

Solution:

Enumerating all equally likely combinations gives Girl Girl, Girl Boy, Boy Girl, and Boy Boy. We are given the information that at least one of them is a girl, therefore the Boy Boy combination is eliminated. All combinations are still equally likely meaning the answer is $\frac{1}{3}$. This is also shown diagrammatically in table IV where 1s indicate valid states.

Table IV: The state space for the easy variant

	Girl	Boy
Girl	1	1
Boy	1	

Enumerating all states as before with the harder variant is far more tedious, but they can be represented easily in a diagram like before as shown in table V. The numerator is given by the number of valid states in the Girl Girl quadrant and the denominator is the total number of valid states, giving a final answer of 13/27.

Table V: The state space for the harder variant

		Girl Mon	Girl Tue	Girl Wed	Girl Thur	Girl Fri	Girl Sat	Girl Sun	Boy Mon	Boy Tue	Boy Wed	Boy Thur	Boy Fri	Boy Sat	Boy Sun
Girl	Mon		1												
Girl	Tue	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Girl	Wed		1												
Girl	Thur		1												
Girl	Fri		1												
Girl	Sat		1												
Girl	Sun		1												
Boy	Mon		1												
Boy	Tue		1												
Boy	Wed		1												
Boy	Thur		1												
Boy	Fri		1												
Boy	Sat		1												
Boy	Sun		1												

Extensions and Comments:

This can also be found more directly with Bayes' theorem which allows us to generalise the additional information known about the chosen girl. In our case the probability of being born on a Tuesday is $\frac{1}{7}$, but we can work with an arbitrary property A with probability x (we assume this having this property to be independent of being born a girl, such as being born on a Tuesday). Bayes' rule is shown below in equation (10) where G_A denotes a girl with property A .

$$P(GG \mid G_A) = \frac{P(G_A \mid GG)P(GG)}{P(G_A)} \quad (10)$$

The first probability in the numerator is given by the complement of two girls independently not having property A given there are two girls which is $1 - (1 - x)^2$. The law of total probability can be used to write the denominator as shown in equation (11a). We have a common factor of $\frac{1}{4}$ as $P(GG) = P(GB) = P(BG) = P(BB) = \frac{1}{4}$. The first of the remaining terms is what was found previously and the last term is immediately seen to be 0. As there is only one girl in each of the middle terms, each is simply the probability of having property A which is x . Putting this together gives equation (11b).

$$\begin{aligned} P(G_A) &= P(G_A \mid GG)P(GG) + \\ &\quad P(G_A \mid GB)P(GB) + \\ &\quad P(G_A \mid BG)P(BG) + \\ &\quad P(G_A \mid BB)P(BB) \end{aligned} \quad (11a)$$

$$P(GG \mid G_A) = \frac{(1 - (1 - x)^2) \cdot x^2}{\frac{1}{4}(1 - (1 - x)^2 + x + x + 0)} = \frac{2 - x}{4 - x} \quad (11b)$$

4 GEOMETRY

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4.1 The Carpet Cutting Problem

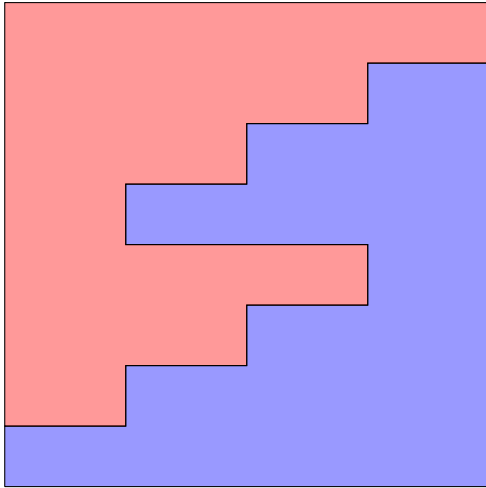
This problem features a rather contrived example that only mathematicians would not question. I found this on the puzzles stack exchange, and most of the time that I spent solving it was used making a tool in Desmos to help test solutions. Once the tool was made, the logical solving path became much clearer, and I would recommend all trying to solve this to use the Desmos or a similar tool. Here is one I made (spoiler free): <https://www.desmos.com/calculator/clrourbpeo>.

A carpet is needed for a 7 metre by 10 metre room. There is a 6 metre by 1 metre section of carpet that cannot be cut, and an 8 metre by 8 metre square of carpet that can only be cut once. This cut can be as complicated as one desires, although cutting to the edge and then back in to the carpet again at the same point counts as two cuts and is not allowed. This includes diagonal, or even curved cuts.

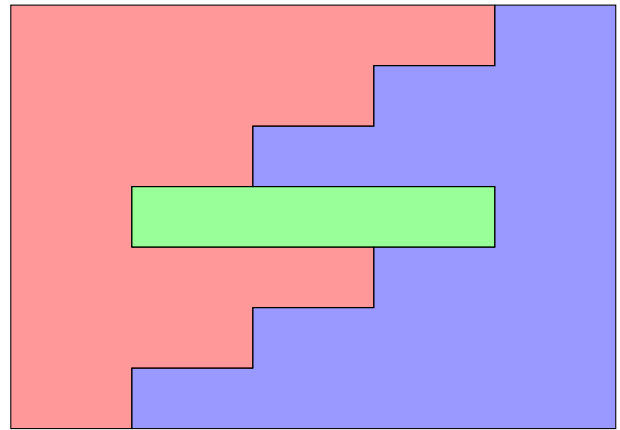
Hints:

1. The cut is symmetrical.
2. The cut uses only orthogonal straight edges.

Solution:



(a) Original 8×8



(b) Arranged into 7×10

Extensions and Comments:

4.2 Semicircle and Quarter Circle Inscribed in Square

Hints:

- 1.

Solution:

Extensions and Comments:

4.3 Painting a Cube

During my PGCE at King's College London we were given a simple 10 minute exercise given and then asked to look for extensions. Together with a friend, Amy, we found this extension that the course leader probably was not expecting. Having something to write with will help significantly for this problem, and the initial problem lends itself very well to an explorative approach.

A $4 \times 4 \times 4$ arrangement of cubees form a larger cube. The larger cube is then painted. The original question was to find how many cubees have exactly 1, 2, 3, 4, 5, and 6 faces painted. Find a formula for how many cubees have exactly k painted faces when an N^m hypercube is painted.

Hints:

- 1.

Solution:

Extensions and Comments:

4.4 Triangles In A Cube

My dad gave me this problem as a simple problem with a fast answer and said it could be solved within 10 seconds, although I found a way of solving this even faster. I would have expected almost anyone to be able to solve this, although I have been surprised that normal people have pretty poor geometric reasoning abilities. I rate this problem a 1/10 in difficulty.

How many equilateral triangles can be formed by connecting three vertices of a cube?

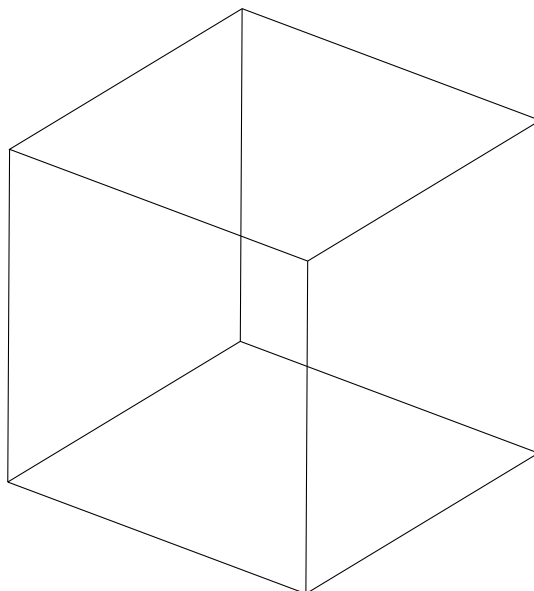
Hints:

1. Connecting any three distinct vertices on the same face will form a right-angled triangle.
2. Equilateral triangles have rotational symmetry of order 3 - where on a cube do you find such a symmetry?

Solution:

Place three points overlapping at a vertex of the cube. There are three edges connected to this vertex, and we move the points along each of the edges. If all points move along their edge at the same speed then they will form an equilateral triangle by symmetry. Eventually they hit the vertex at the end of the edge, and thus we have found three vertices that form an equilateral triangle. This gives one triangle per vertex, therefore the final answer is 8. We have not shown that these are the only such triangles, although it is not hard to convince yourself by visualising it. The next method proves that these are all the equilateral triangles however.

There are 24 orientation-preserving symmetries of the vertices of a cube, and these form the O group. The vertices of an equilateral triangle have 3 orientation-preserving symmetries and form the C_3 group. The quotient set O/C_3 has size $\left| \frac{O}{C_3} \right| = \frac{|O|}{|C_3|} = \frac{24}{3} = 8$, and this is the number of triples of vertices with threefold rotational symmetry, in other words, the equilateral triangles.

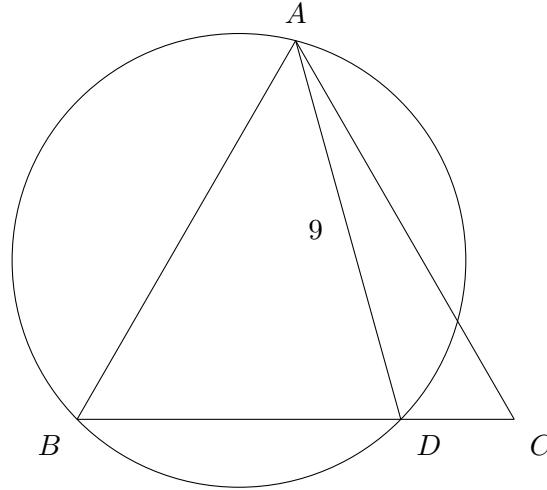


Extensions and Comments:

4.5 Almost Inscribed Triangle In Circle

I found this from Catriona Agg's collection of geometry problems and it appealed to me as it seemed very minimal. The first solution was found by me and the second simpler solution was found by my dad. I rate this problem a 4 out of 10 in terms of difficulty.

The triangle ABC is equilateral and $|AD| = 9$. Find the area of the circle.



Hints 1:

1. The triangle ADC contains segment AC which is equal in length to segment AB . Use this to construct a congruent triangle.
2. Prove that the added point of this constructed triangle lies on the circle using the converse of the alternating quadrilateral theorem.
3. Prove that the triangle formed by A , B , and the added point is equilateral.

Solution 1:

Noting that $|AB| = |AC|$, we construct triangle EBA to be congruent to triangle DCA as shown highlighted in figure 4a. Our aim is to show that the point added to form this triangle, E , lies on the circle. Defining $\alpha = \angle BAE$ and $\beta = \angle DAB$, we find $\angle DAE$ by equation (12).

$$\angle DAE = \angle DAB + \angle BAE = \beta + \alpha = \alpha + \beta = \angle CAD + \angle DAB = 60^\circ \quad (12)$$

By congruence of triangles EBA and DCA , $\angle EBA = \angle DCA = 60^\circ$, and thus $\angle EBD = 120^\circ$. Therefore $\angle EBD$ and $\angle DAE$, opposite angles of quadrilateral $AEBD$, add to 180° . As $\angle AEB$ and $\angle ADC$ are equal by congruence of triangles EBA and DCA and $\angle BDA$ and $\angle ADC$ make the straight line BDC , we deduce that opposite angles of quadrilateral $AEBD$ ($\angle AEB$ and $\angle BDA$) add to 180° as well. Therefore quadrilateral $AEBD$ is a cyclic quadrilateral and thus point E lies on the circle.

By definition $|AD| = |AE|$ which makes triangle DAE isosceles. As shown before, $\angle DAE = 60^\circ$, therefore $\angle AED + \angle EDA = 120^\circ$. As these angles are equal from triangle DAE being isosceles they must both be 60° , and thus triangle DAE is in fact equilateral. By rotational symmetry the centre of of the triangle must be the centre of the circle. With the simple construction given in figure 5b and basic trigonometry we see that the area of the circle is 27π as shown in equation (13).

$$r = \frac{4.5}{\sin(60^\circ)} = \frac{4.5}{\frac{\sqrt{3}}{2}} = 3\sqrt{3} \implies \text{Area} = \pi r^2 = 27\pi \quad (13)$$

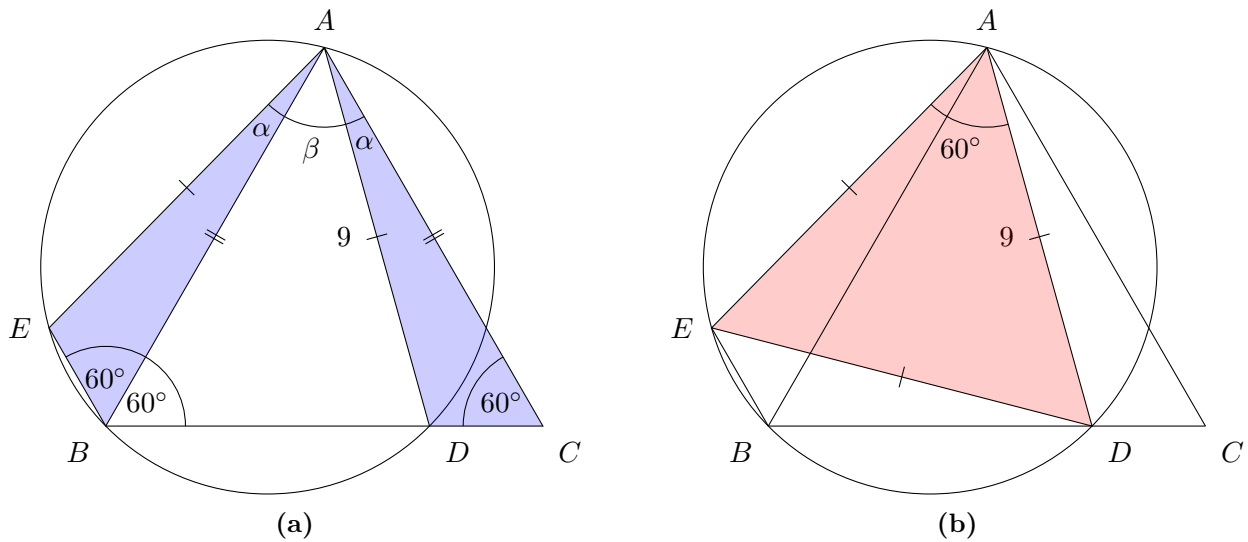


Figure 4: Solution 1

Hints 2:

1. Find an angle at the centre of the circle.
2. Draw a kite connecting the circle centre, A , B , and D .
3. The angle subtended by an arc at the circle centre is double the angle subtended by the same arc at the circle edge.

Solution 2:

Connect the circle centre, O , to points A and D to form a kite, as shown in figure 5a below. As the angle subtended by an arc at the centre is double the angle subtended at the circle edge we get $\angle DOA = 2 \cdot \angle DBA = 120^\circ$ and the solution continues identically to solution 1.

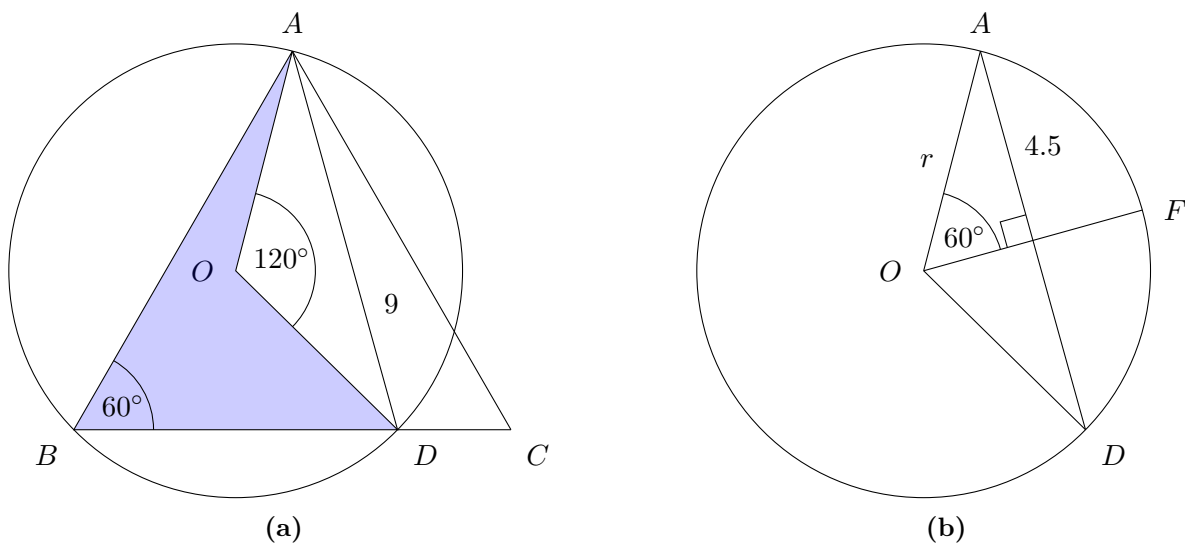


Figure 5: Solution 2 and the common method for finding the circle area

Extensions and Comments:

I tried solving this while on the train and was not bothered to get my whiteboard out. I managed to find and prove the cyclic quadrilateral, but did not spot the isocoles triangle.

5 PHYSICS

5.1 The Ball and the Boat Problem	51
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5.1 The Ball and the Boat Problem

This is a classic problem, but I first did a proper analysis of it when I was looking for Oxbridge mock interview questions and my friend, Dami from WSP, suggested it. This requires only very basic understanding of physics and does not need any particular tricks or special insights to solve, but is nevertheless quite pleasing. I rate this problem a 2 out of 10 in terms of difficulty.

Suppose there is a small boat in a pool with a person and a ball onboard. The person throws the ball overboard. What happens to the water level in the following cases?

1. The ball sinks to the bottom.
2. The ball floats on the surface.
3. The ball floats but it is then tied to the bottom of the pool.
4. The ball floats but the person in the boat forces it underwater so it is fully submerged.

Hints:

1. Remember Archimedes' principle - the buoyant force acting on an object is equal to the weight of fluid it displaces, and acts in the upwards direction.
2. Consider extreme cases - a very dense ball and a ball with very low density.
3. Identify the different systems and the forces acting on them.

Solution:

The key to this problem is understanding difference between buoyant forces caused by the mass of the ball and the volume of the ball. When the ball is in the boat, the buoyant force is due to the weight of the ball and is independent of the volume. When the ball is in the water, the buoyant force is due to the volume of the ball and is independent of the weight. The forces in the original situation are shown in figure 6, and the four cases are shown in figure 7.

When the ball is in the boat, the additional weight pushes the boat deeper into the water, meaning a greater volume of the boat is under the water line. This means a greater volume of water is displaced, and thus a greater weight of water is displaced, and therefore a larger buoyant force. This force is a function only of the weight of the ball as changing the volume of the ball only displaces air inside the boat. When the ball is fully submerged, the volume of water displaced is equal to the volume of the ball and therefore the weight of the ball is irrelevant. If the ball is floating, then by Archimedes' principle the weight of the water displaced by the volume of the ball is equal to the weight of the ball.

This understanding can be used to construct a force balance. When the ball is in the boat, the boat and the ball are treated together as a single system. The two forces are the weight of the ball and the boat combined acting downwards, and an equal and opposite buoyant force acting upwards. When the ball is fully submerged, the ball and the boat need to be treated as separate systems. As before, the boat has two forces acting on it, the weight of the boat and the equal and opposite buoyant force. The case for the ball is different however. The ball has a weight acting downwards, but now there are two forces acting upwards. The buoyant force is equal to the weight of water displaced by the ball and there is also a reaction force from the bottom of the pool stopping the ball falling through the floor.

In both cases, the systems are in equilibrium and the total downwards forces are equal, meaning the upwards forces must be equal in both cases as well. As there is a contact force when the ball is submerged, the total buoyant force must be less than when the ball is in the boat. The buoyant force is equal to the weight of the water displaced, and as the water is of constant density, this is proportional to the volume of water displaced. This means less water is displaced when the ball is submerged compared to being in the boat, and thus the water level is lower. We could also see this by noting the weight of the ball must be greater than the weight of the water it displaces because it is denser than water.

Similar reasoning can be applied in the other parts of the question. If the ball floats on the surface or has a density equal to that of water there is no contact force. The total system weight is equal to the total system buoyant force in both cases, meaning the water level stays the same. If the ball has a density less than water but is forced under the surface anyway, the buoyant force is more than the weight. Instead of a contact force acting upwards, a tension in the rope acts downwards to maintain equilibrium. The greater total buoyant force means more water is displaced, giving a higher water level.

The final part is almost identical to the second part. The total buoyant force is equal to the total weight as there are no other forces acting on the system and it is in equilibrium. As before this leads to the water level staying the same. Understanding the individual forces on the ball and boat is still interesting however. The buoyant force on the ball is split into two parts, one equal to the weight of the ball, and the remaining buoyancy is additional buoyancy. As the weight of the boat is smaller than the boat and the ball combined, the boat will float higher in the water due to a smaller volume needing to be displaced. This loss in buoyancy is exactly equal to the additional buoyancy from the ball. Compared to the second part of the question, the boat sits higher in the water and the ball sits lower, with the total water displaced remaining exactly equal.

Extensions and Comments:

This is a problem where thinking of extreme cases helps elucidate the principles. Consider a very tiny ball of constant size and large mass. The difference in water level when the ball is in the water or completely removed from the system is negligible due to the small size of the ball. Increasing the weight makes no difference at all. If the ball was put into the boat however, the boat would sit much lower in the water. Increasing the weight of the ball would cause it to sit even lower, and clearly the volume of water displaced increases. A very light ball with a large varying volume demonstrates part three nicely. This makes very little difference to how deep the boat sits in the water, and is independent of the volume. When increasing the volume when it is submerged, the volume of water displaced increases, and the water level clearly increases.

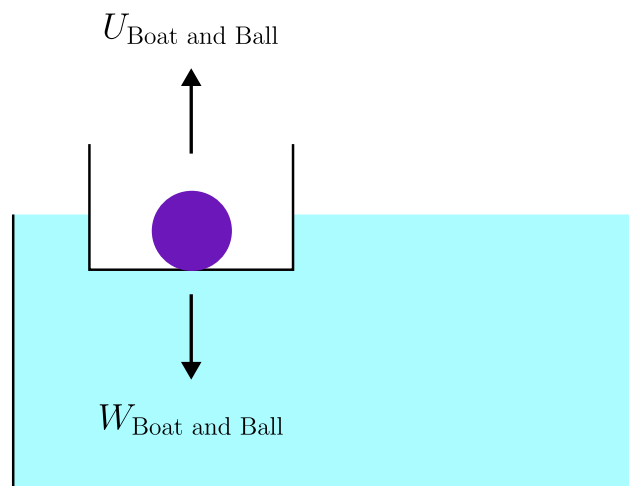


Figure 6: The forces acting on the original system

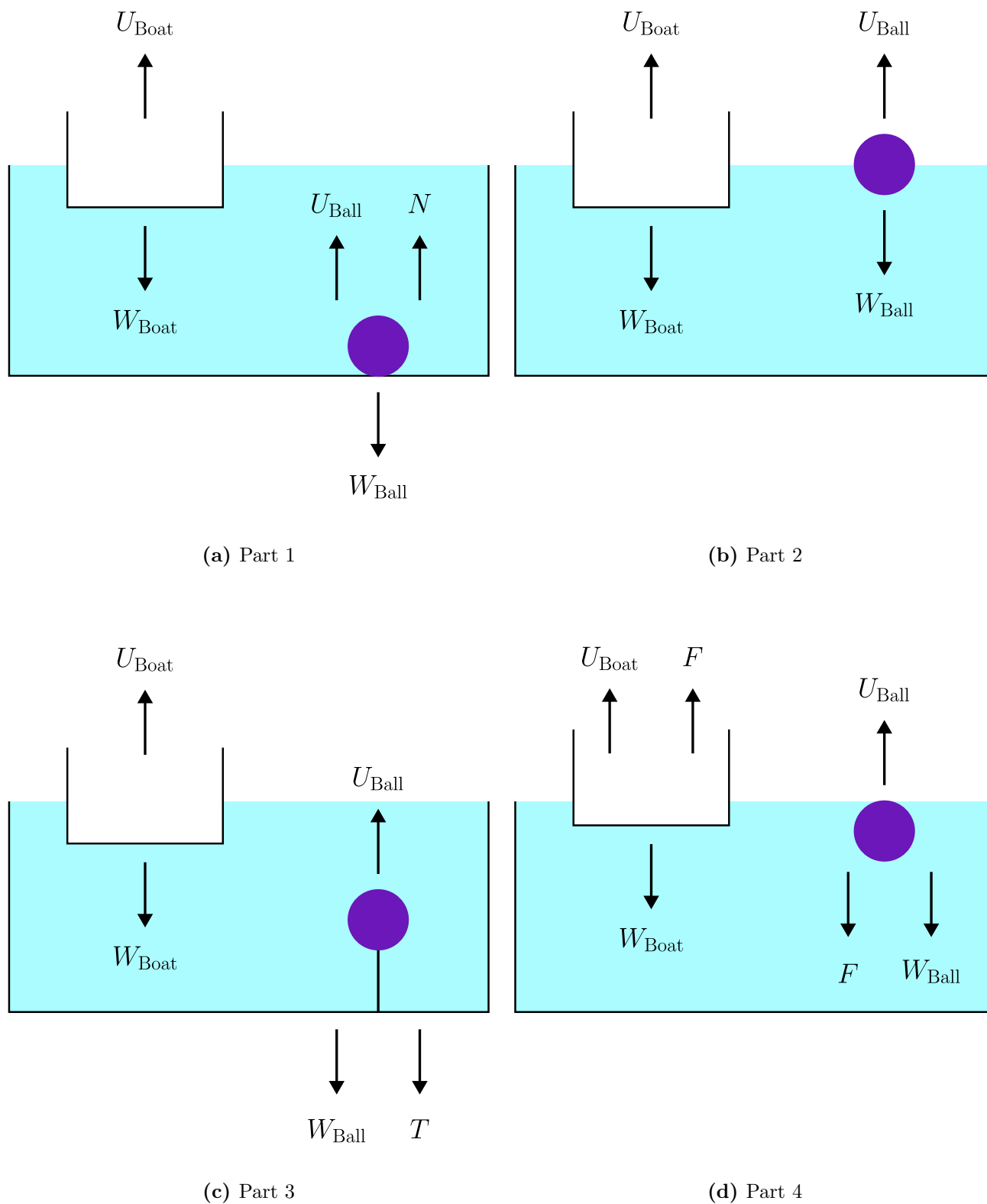


Figure 7: The forces acting on the ball and the boat in each of the four parts

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- [1] Kseniya Garaschuk and Andy Liu. *Grade Five Competition from the Leningrad Mathematical Olympiad*. Springer, 2020.
- [2] Alan Lauder. Part A Linear Algebra MT 2019 Sheet 1 of 4. *Oxford Mathematics Undergraduate Course*, 2019.